

Coding with Kanak

Data Warehousing & Strategic Information Management

Unit 1



Data Warehousing: Unit-1

1.1 Data Warehouse

A data warehouse is just a bigger database. It is a treasure chest filled with historical data. This data comes from different business applications and outside sources.

1.2 Data Warehousing

Data warehousing is the process of collecting, integrating, storing, and managing data from multiple sources in a central repository.

A data warehouse works like a main hub. It collects different sources of data, cleans it up, and organizes it well. Data warehousing is not just about gathering information. It's also about changing it. It turns into something structured and ready for analysis. It enables organizations to organize large volumes of historical data for efficient querying, analysis, and reporting.

The main goal of data warehousing is to support decision-making by providing clean, consistent, and timely access to data.

1.3 Need for Data Warehousing

- **Handling Large Data Volumes** : Traditional databases store limited data (MBs to GBs), while data warehouses are built to handle huge datasets (up to TBs), making it easier to store and analyze long-term historical data.
- **Enhanced Analytics**: Databases handle transactions; data warehouses are optimized for complex analysis and historical insights.
- **Centralized Data Storage** : A data warehouse combines data from multiple sources, giving a single, unified view for better decision-making.
- **Trend Analysis**: Data warehouse allows businesses to analyze trends over time, enabling

them to make strategic decisions based on past performance and predict future outcomes.

- Business Intelligence Support: Data warehouses work with BI tools to give quick access to insights, helping in data-driven decisions and improving efficiency.

1.4 Characteristics of Data Warehousing

A data warehouse has four key characteristics: it is Subject -Oriented, focusing on specific business topics like sales or customers; Integrated, consolidating data from various sources into a unified format; Time-Variant, storing historical data to track trends over time; and Non -Volatile, meaning data is read-only, not changed or deleted after it is loaded.

1. Subject-Oriented:

Data is organized around major subjects of the business, such as customers, products, or sales, rather than around individual applications or operational processes. This allows for comprehensive analysis of a particular business area by bringing together all relevant data from different sources into one place.

2. Integrated:

Data from disparate sources (databases, cloud systems, etc.) is consolidated and transformed into a consistent, unified format. This ensures data is clean, reliable, and compatible, providing a single, trustworthy source of truth for decision-making.

3. Time-Variant:

A data warehouse stores historical data over long periods, often in the form of data snapshots. This enables users to analyze trends, track changes, and gain insights into how the business has performed over time.

4. Non-Volatile:

Once data is loaded into the data warehouse, it is not modified or deleted. This read -only characteristic makes the data stable and reliable, preserving a historical record and ensuring consistency for analytical queries.

1.5 Types of Data Warehouses

The different types of Data Warehouses are:

1. Enterprise Data Warehouse (EDW) : A centralized warehouse that stores data from across the organization for analysis and reporting.

2. Operational Data Store (ODS): Stores real -time operational data used for day -to-day operations, not for deep analytics.
3. Data Mart: A subset of a data warehouse, focusing on a specific business area or department.
4. Cloud Data Warehouse : A data warehouse hosted in the cloud, offering scalability and flexibility.
5. Big Data Warehouse: Designed to store vast amounts of unstructured and structured data for big data analysis.
6. Virtual Data Wareho use: Provides access to data from multiple sources without physically storing it.
7. Hybrid Data Warehouse: Combines on-premises and cloud-based storage to offer flexibility.
8. Real-time Data Warehouse : Designed to handle real -time data streaming and analysis for immediate insights.

1.6 Data Warehousing vs. Data Mining

1.6.1 Data Warehousing:

Data warehousing is the process of collecting, organizing, and managing structured data from multiple sources into a single, central repository is known as data warehousing. It is primarily intended to facilitate and assist business intelligence (BI) and analytics operations.

1.6.2 Data Mining:

Data mining is a process of collecting valuable insights, trends, and patterns from huge datasets. Many different statistical and machine learning techniques are used to find the hidden patterns, relationships, and anomalies within the data.

1.6.3 Difference between Data Warehousing and Data Mining

Data mining is not similar to data warehousing, Data warehousing focuses on managing and storing data, while data mining focuses on analyzing data to extra ct essential actionable insights.

Aspect	Data Warehousing	Data Mining
Purpose	Store and manage structured data	Discover patterns and relationships

Nature of

Data

Historical, structured, integrated data

Uncover hidden patterns in large

datasets

Focus Data storage and retrieval Data analysis and pattern recognition

Techniques

Online Analytical Processing and Extract,

Transform, Load which is OLAP and ETL

Clustering, classification, association

rules

Output Organized data repositories Predictive models, patterns, trends

1.6.4 Characteristics Differences – Data Warehousing vs. Data Mining

- Purpose:

For reporting and analysis data warehousing majorly focus on storing data as well as managing structured data. While the goal of data mining is to find links and patterns in the data so that choices or predictions can be made.

- Nature of Data:

Data warehousing deals with structured and historical data which is integrated from multiple sources but data mining looks after the hidden pattern in the big data set which includes structured as well as unstructured data.

- Focus:

The focus of data warehousing is on data retrieval and storage, guaranteeing data availability and consistency for analysis. While the data mining concentrates on pattern identification and data analysis with the goal of extracting useful insights from the data.

- Techniques:

Data warehousing employs techniques such as ETL and OLAP for data integration and analysis, whereas data mining utilizes algorithms like clustering, classification, and

association rule mining to uncover patterns and relationships within the data.

- Output:

The output of data warehousing is organized data repositories, providing a unified view of data for reporting and analysis purposes. In contrast, data mining produces predictive models, patterns, trends, or anomalies discovered from the data.

1.6.5 How does Data Warehousing and Data Mining Work Together

While both have served different purposes, they are closely interconnected and often work together to leverage the full potential of data. Data warehousing provides the foundation by storing and organizing data in a centralized repository, making it accessible for analysis. Data mining then utilizes this data to uncover valuable insights, which can be used to make informed decisions, improve business processes, and gain a competitive edge. For example, a retail company may use a data warehouse to consolidate sales data from various stores and online channels. Researchers can then apply data mining techniques to analyze this data and identify customer purchasing patterns, preferences, and trends. These insights can inform marketing strategies, inventory management, and product recommendations, ultimately driving business growth and profitability.

1.7 Data warehouse vs. Data Lake vs. Data Mart vs. Database

Data warehouse vs. Data Lake

Both data warehouses and data lakes are used for storing Big Data and complement each other, but they are very different storage systems. A data warehouse stores data that has been formatted for a specific purpose, whereas a data lake stores data in its raw, unprocessed state – the purpose of which has not yet been defined.

When raw data stored in a lake is needed to answer a business question, it can be extracted, cleaned, transformed, and used in a data warehouse for analysis.

1.8 Data warehouse vs. Data mart

A data mart is a subsection of a data warehouse, partitioned specifically for a department or line of business – like sales, marketing, or finance. While a data warehouse serves as the central data store for an entire company, a data mart serves relevant data to a select group of users. This

simplifies data access, speeds up analysis, and gives them control over their own data. Multiple data marts are often deployed within a data warehouse.

1.9 Data Warehouse vs. Database

Database Data Warehouse

A common Database is based on operational or transactional processing. Each operation is an indivisible transaction.

A data Warehouse is based on analytical processing.

Generally, a Database stores current and up-to-date data which is used for daily operations.

A Data Warehouse maintains historical data over time. Historical data is the data kept over years and can be used for trend analysis, make future predictions and decision support.

A database is generally application specific.

Example: A database stores related data, such as the student details in a school.

A Data Warehouse is integrated generally at the organization level, by combining data from different databases.

Example: A data warehouse integrates the data from one or more databases, so that analysis can be done to get results, such as the best performing school in a city.

Constructing a Database is not so expensive. Constructing a Data Warehouse can be expensive.

1.10 Operational versus analytical databases

1.10.1 Operational Database

Operational data, as the name implies, refers to the data generated by our organization's daily operations. Example customer information, purchase orders, sales records, inventory lists, and HR databases.

Operational data typically is stored in a consistent format, as it is a reliable source for obtaining the most current information on various business departments.

Operational data can be classified into different categories depending on our organization's departments and style. We can create a business operational data segment that has relevant datasets for your business processes, customers, sales, or purchase data.

It is typically Online Transaction Processing (OLTP) systems. These databases are highly optimized for fast, frequent, and small transactions, such as inserts, updates, and deletes. They prioritize data integrity, concurrency, and immediate availability for specific, well-defined business use cases.

1.10.2 Analytical Database

Analytical data refers to the datasets that are ready for analysis after being consolidated, standardized, and transformed.

Once we have brought together all our operational data from different sources, we must verify, categorize, and standardize each section. Our datasets can be further divided into different specialized categories like customer segmentation, sales data, and more.

Analytical data is often utilized for making strategic business decisions. We perform various analyses on it to determine demand patterns, the latest trends, consumer behavior shifts, and more.

It is typically Online Analytical Processing (OLAP) systems, designed for efficient querying and analysis of large volumes of historical data. They prioritize query performance for complex analytical tasks over real-time transaction processing.

1.11 Trends and Planning in Data Warehousing

In data warehousing, trends are the evolving technologies and methodologies shaping the field,

such as cloud adoption, real-time analytics, and AI/ML integration, while planning involves the strategic process of defining a data warehouse's scope, goals, and architecture to meet business needs, including data modeling and integration with new technologies.

1.11.1 Trends in Data Warehousing

- Cloud Adoption:

Cloud-based data warehouses are becoming the standard, offering scalability, accessibility, and cost-efficiency compared to traditional on-premises systems.

- Lakehouse Architecture:

A hybrid approach combining the flexibility of data lakes for structured and unstructured data with the structured management of data warehouses.

- Real-Time Analytics:

The shift from batch processing to real-time data ingestion and analysis provides immediate insights, moving from a luxury to a necessity for businesses.

- AI and Machine Learning:

Integration of AI/ML capabilities enables predictive modeling, automated insights discovery, and advanced analytics.

- Data Mesh and Data Fabric:

Emerging architectures that decentralize data management and ownership while emphasizing data interoperability and exchange, according to Acceldata.

1.11.2 Planning a Data Warehouse

- Identify Data Needs:

Conduct a comprehensive assessment of business goals, data requirements, and objectives to understand what data is needed and for what purpose.

- Design the Data Model:

Structure the data logically, choosing a schema (like a star or snowflake) and defining relationships between entities (e.g., customers, products) to ensure efficient storage and retrieval.

- Prioritize Optimizations:

Focus on performance, scalability, and usability enhancements to make the data

warehouse effective and responsive.

- **Select Technology:**

Choose cloud platforms, Lakehouse architectures, or other technologies that align with your organization's needs for cost, performance, and scalability.

- **Plan for Integration:**

Ensure the data warehouse can seamlessly integrate with existing external data sources, leveraging APIs for efficient data capture.

1.12 Strategic Information Management

Strategic Information Management (SIM) is the systematic planning, management, and use of an organization's information resources to achieve its long-term business objectives and gain a competitive advantage.

Need Strategic Information Management

Strategic Information Management is necessary for organizations to gain competitive advantages, make informed decisions, achieve long-term goals, and improve operational efficiency in today's dynamic business environment. It involves aligning information systems with business strategies to process and analyze information, identify opportunities and risks, monitor performance, and foster innovation.

Management needs information to formulate the business strategies, establish goals, set objectives, and monitor results

Key Concepts

- **Strategic Alignment:**

Ensuring that information systems and technology initiatives directly support the organization's overall business strategy and goals.

- **Competitive Advantage:**

Using information and technology to gain an edge over competitors by providing better products, services, or customer experiences.

- **Data-Driven Decision-Making:**

Leveraging data analytics and metrics to gain insights, identify opportunities for

improvement, and make informed strategic choices.

- Information Governance:

The overall framework for how information is collected, stored, processed, and used within an organization, ensuring data quality, security, and compliance.

The Strategic Information Management Process

- Goal-Setting/Identification:

Clearly define the organization's long-term vision, mission, and strategic objectives to guide information management efforts.

- Analysis:

Conduct a comprehensive analysis of the internal and external business environment using tools like SWOT to understand strengths, weaknesses, opportunities, and threats.

- Strategy Formulation:

Develop an action plan and identify the specific information systems and technologies needed to achieve the established goals, aligning them with the business plan.

- Strategy Implementation:

Put the plan into action, deploying the chosen information systems and resources, with clear communication and strong leadership.

- Evaluation and Control:

Continuously monitor the performance of information systems, measure success against established metrics, and make necessary adjustments to the strategy.

Benefits of Strategic Information Management

- Improved Operational Efficiency:

Streamlining processes and reducing costs through better data management.

- Enhanced Decision-Making:

Providing timely and accurate information for better strategic and operational decisions.

- Increased Competitiveness:

Differentiating the organization in the marketplace and attracting more customers.

- Adaptability to Change:

Enabling organizations to effectively respond to technological advances and market shifts

1.13 Decision Support System (DSS)

A decision support system (DSS) is a computer-based tool that helps individuals and organizations make better decisions by integrating data, models, and analytics to provide actionable insights and support strategic planning. It allows users to explore "what-if" scenarios and interact with vast amounts of data, offering flexibility and user-friendliness to solve complex, often semi-structured problems.

Components of a DSS

The primary components of a DSS are a database, a model base, and a user interface, enabling users to query information, run models, and visualize outcomes.

- Database Management Subsystem:

This component collects and stores data from various internal and external sources, such as sales data, customer information, and market trends. It acts as the core repository of information that the decision-maker uses.

- Model Management Subsystem:

This component holds the mathematical and analytical models (e.g., financial, forecasting, simulation models) that process the data. It helps in analyzing data, exploring different scenarios, and simulating potential outcomes to support decision-making.

- User Interface:

This is the point of interaction between the user and the DSS, providing a graphical interface for users to input data, navigate the system, and view results. It makes the system accessible and user-friendly for non-technical individuals.

- Knowledge Base (Optional but increasingly common):

Some frameworks, or newer versions of a DSS, also include a knowledge base that stores rules and procedures to provide advice or automated recommendations. It helps the decision-maker by providing expert knowledge and context for complex problems.

Key Characteristics

- **Interactive:** Users can actively engage with the system, query data, and manipulate models.
- **Data Integration:** It combines information from various internal and external sources.
- **Model-Based:** It incorporates mathematical and statistical models for analysis.
- **User-Friendly Interface:** The system presents information in an easy-to-understand.
- **Support for Unstructured Problems:** DSS is particularly effective for complex, semi-structured, and unstructured decisions where simple rules aren't sufficient.

How a DSS Works

- **Data Gathering and Analysis:**

The system collects and analyzes large volumes of data from multiple sources.

- **Model Application:**

Mathematical and statistical models are used to analyze the data and simulate outcomes.

- **Information Presentation:**

Insights are presented to decision-makers in a clear, accessible format, such as reports or graphical representations.

- **User Interaction:**

Decision-makers can interact with the system to explore various scenarios, ask "what-if" questions, and adjust parameters to see the potential results.

Benefits of Using a DSS

- **Improved Decision Quality:**

By providing data-driven insights, a DSS helps to reduce uncertainty and improve the quality of decisions.

- **Enhanced Efficiency:**

It automates the process of data analysis and report generation, saving time and increasing efficiency.

- **Competitive Advantage:**

DSS helps organizations gain a competitive edge by enabling faster, more informed strategic and operational choices.

- **Better Problem-Solving:**

It provides tools and frameworks to effectively address complex operational and strategic

issues.

Application of DSS

- Sales and Marketing:

DSS can project revenue, analyze sales trends, and support inventory management.

- Financial Planning:

These systems assist with financial modeling, budgeting, and scenario analysis to evaluate different financial strategies and market conditions.

- Operations:

DSS supports operations management by synthesizing data into actionable intelligence, such as optimizing supply chains.

- Healthcare

- Diagnosis and Treatment:

DSS can analyze patient data, medical records, and clinical guidelines to aid in diagnosis and treatment planning.

- Patient Safety:

By rapidly analyzing metrics and running tests, DSS can help improve patient safety and potentially reduce medical errors.

1.14 Online Transaction Processing (OLTP) & Online Analytical Processing (OLAP)

OLTP (Online Transaction Processing) systems manage day -to-day, high -volume, real -time transactions like purchases and withdrawals, prioritizing data integrity and fast, simple operations.

- Purpose: To handle a large number of concurrent, simple transactions in real-time.
- Focus: High-speed, accurate transaction processing and data integrity.
- Data: Stores and processes large volumes of current transactional data.
- Operations: Primarily involves Insert, Update, and Delete (IUD) operations.
- Examples: ATM banking, e-commerce checkout, and online purchases.
- Users: Typically clerks, managers, and operational staff.

OLAP (Online Analytical Processing) systems, on the other hand , focus on analyzing large

historical datasets to provide business insights through complex queries and reporting, enabling data-driven decision-making.

- Purpose: To analyze large volumes of historical data from various perspectives to derive meaningful business insights and trends.
- Focus: Complex queries, advanced data analysis, and data mining.
- Data: Data is often aggregated and stored in multidimensional databases or data warehouses.
- Operations: Focuses on Read-only operations, though data is updated in scheduled batches.
- Examples: Business intelligence, sales analysis, financial reporting, and supply chain management.
- Users: Business analysts, CEOs, and decision-makers.

Key Differences in a Nutshell

- Function: OLTP handles transactions, while OLAP performs analysis.
- Data Volume: OLTP deals with smaller, individual transactions; OLAP works with vast datasets.
- Speed: OLTP requires fast response times for simple queries; OLAP involves complex queries that can take longer.
- Data Model: OLTP databases are often application-focused, whereas OLAP systems use multidimensional models

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